Alban Pasquelin

Software Architect & Tech Lead | React Native · React · TypeScript · Three.js

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PROFILE

Twenty years of product engineering across web, mobile, desktop and embedded environments. I take on front-end and cross-platform architecture, heavy migrations that must not interrupt service, and real-time 3D. Co-founder and CTO of GoSecure since 2017, CTO of RetroRoads. What I am building up now is the **industrialisation** of products backed by generative models: asset management, performance, error recovery, persistent state.

EXPERIENCE

Co-founder & CTO · GoSecure

2017 → present

A public-safety platform: mobile app plus a real-time SaaS console.

- Whole-system architecture, from mobile to the real-time backend, in production for nine years
- Geolocated alerts with photo or video; a console that receives and coordinates response
- API integration with the systems clients already run; reporting dashboards
- Deployed with companies, local authorities and transport operators
- Selected by the 2026 Propulse programme, French Transport Innovation Agency

`React Native · React · Node.js · real time · SaaS`

CTO · RetroRoads

May 2026 → present

A consumer app for classic cars. Currently in private beta.

- Architecture of all three parts: GraphQL API, admin panel, mobile app
- The GraphQL schema is the source of truth; clients consume a generated mirror
- One cross-cutting visibility model in a single service; content translated into twenty languages
- Expo mobile app with native maps, GPS route recording, CarPlay and Android Auto

`Express 5 · Apollo · TypeGraphQL · TypeORM/MySQL · Redis · RabbitMQ · Expo SDK 57`

React front-end developer, internal tool rebuild · Ardian

Jul 2025 → Jan 2026

Investment firm. Rebuild of a strategic internal tool.

- Stack and UX modernisation; contribution to the front-end architecture
- Design system, Figma implementation, REST API integration

`React · Tailwind CSS · DaisyUI · REST`

Edito CMS rebuild lead · e-TF1

Jan 2024 → Jul 2025

- Led the full rebuild of the editorial CMS, migrating the front-end from Vue.js to React
- Moved to end-to-end TypeScript, sharply reducing production regressions
- GraphQL for client-server exchange, schema validation, atomic components
- Data integrity preserved throughout; load times cut by more than 30%

`React · TypeScript · GraphQL · Apollo · Tailwind CSS`

IPTV architect, MyTF1 across all platforms · e-TF1

Jan 2021 → Jan 2024

- Rebuilt MyTF1 across every French operator set-top box from a single codebase
- Shipped to Android TV, WebOS and WPE, each with its own hardware constraints
- Automated white-label site generation for the group's thematic channels
- SCTE-35 dynamic ad insertion integrated with the Freewheel stack

`React Native · React Native Web · Android TV · WebOS · WPE · SCTE-35`

Senior front-end developer · Innso

Mar 2019 → Dec 2020

Customer-relations platform rebuild; front-end architecture and a chatbot design studio.

`Vue.js · Vuex · Vuetify · cross-platform iOS/Android`

Head of TV division, then head of TV & product design advisor · Molotov TV — Feb 2015 → Dec 2016

Launch of the service. First head of the TV division, from the startup's early phase to release.

- Built the TV division: team, product backlog, development methodology
- Designed and built a DRM-capable HTML5 video player for secure delivery
- Connected-TV apps (Apple TV, LG, Samsung), plus iPhone, iPad and desktop
- Owned the experience and the interface across every platform
- **Apple App of the Year 2016**

`React · JavaScript ES6/ES7 · HTML5 video · DRM · Apple TV · Smart TV`

Earlier work — 2005 → 2019

Infopro Digital (real-time webinar platform, Vue.js and OpenTok) · Arte (Apple TV app, tvOS) · Equidia (information system rebuild, via DonkeyCode) · Staffmatch · Amplement (R&D and product strategy director) · Business Lab (lead developer, then technical project manager, team of ten) · Peugeot · Leroy Merlin · E.Leclerc (price comparator)

PERSONAL PROJECTS

AI Desktop Studio — 2026

A generative creation studio on the desktop. Seven workspaces, a locally indexed SQLite catalogue, a canvas image editor, a video timeline that actually decodes, generated textures and skyboxes applied inside the 3D viewport. 310 MCP actions: the whole UI is drivable by an agent.

`Electron · React 19 · TypeScript · three.js · PixiJS · SQLite`

Trillion3D — 2026

Game-engine virtualized geometry for the web: a native Rust compiler cuts models into 128-triangle clusters, a WebGPU engine streams them under a fixed memory budget, with GPU culling and temporal antialiasing. Every number is proven on a bench.

`Rust · TypeScript · WebGPU · WebGL2 · WebAssembly`

map3d — 2026

A real-time 3D mapping library for React. Photorealistic globe down to street level, pedestrian mode, a geospatial drawing editor with GeoJSON import and export, typed themes.

`React 19 · Three.js · Google Photorealistic 3D Tiles · WebGL`

panels — 2026

The chassis of a tool application, released under MIT: icon rails, resizable zones, a free centre taking either tabbed documents or an application route. 8 kB gzipped, zero dependencies.

`React 19 · TypeScript · MIT · npm @pasquelin/panels`

SKILLS

Front & cross-platform	React, React Native, React Native Web, Expo, Vue.js, strict TypeScript, Tailwind CSS
Real-time 3D	Three.js, WebGL, WebGPU, PixiJS, Google 3D Tiles, rendering pipelines
Desktop	Electron, SQLite, packaging and updates
Backend & data	Node.js, Express, GraphQL, Apollo, TypeORM, MySQL, Redis, RabbitMQ, Docker, Keycloak
Constrained targets	Android TV, WebOS, WPE, operator set-top boxes, CarPlay, Android Auto, SCTE-35
Practices	Software architecture, design systems, zero-downtime migrations, CI/CD, code review, growing a team

AWARDS

- **Apple**, App of the Year 2016
- **TOPCOM d'Or 2013**, peugeot.com
- **TOPCOM d'Or 2012**, Peugeot 4008
- **Grand prix Stratégies du digital**
- **Propulse 2026**, French Transport Innovation Agency (GoSecure)
- Published in *PHP Solutions*, January 2009

EDUCATION

Accredited degree in graphic and multimedia arts (Orfiale, 2002). CFAcom.

LANGUAGES

French: native. English: full professional proficiency.

Availability: Paris and Île-de-France (on site or hybrid), French-speaking Switzerland, remote across Europe. **Swiss invoicing already in place**, so a client in Geneva has no administrative friction to handle. **Rate**: €750–850 per day in France, CHF 1,000–1,150 in Switzerland.